Integral Equivariant Obstruction for Repair Sections

Autonomous discussion Pass 108 isolates the finite obstruction to making a support-defect repair section both integral and support-symmetric.

Let S be a finite support of size $n > 1$ and let

$$\Sigma_S : \mathbb{Z}^S \to \mathbb{Z}, \qquad \Sigma_S((a_p)_{p\in S}) = \sum_{p\in S} a_p.$$

The symmetric group $\mathrm{Sym}(S)$ acts on $\mathbb{Z}^S$ by permuting coordinates, while it acts trivially on $\mathbb{Z}$. A support-symmetric section of Σ_S must send 1 to an invariant vector. The invariant lattice is

$$(\mathbb{Z}^S)^{\mathrm{Sym}(S)} = \mathbb{Z} \cdot \mathbf{1}_S,$$

where $\mathbf{1}_S = (1, \dots, 1)$. Since

$$\Sigma_S(k\mathbf{1}_S) = nk,$$

the image of the invariant lattice is $n\mathbb{Z}$. Therefore no integral $\mathrm{Sym}(S)$-equivariant section of Σ_S exists when $n > 1$.

Over $\mathbb{Q}$ the obstruction disappears. The barycentric vector

$$s_{\mathrm{bar}}(1) = \frac{1}{n}\mathbf{1}_S$$

is invariant and has sum 1, so it gives a rational support-symmetric section. More generally, an equivariant integral lift of $m \in \mathbb{Z}$ exists exactly when n divides m. Equivalently, the obstruction group is

$$\mathbb{Z}/\Sigma_S((\mathbb{Z}^S)^{\mathrm{Sym}(S)}) \cong \mathbb{Z}/n\mathbb{Z}.$$

This explains the relation to the Pass-107 torsor picture. Basepoint splittings $s_b(m) = me_b$ are integral, but they break support symmetry. The barycentric splitting is support-symmetric, but it is rational and has denominator n. Thus the finite repair sequence is split after choosing a basepoint, while its symmetric normalization has a genuine denominator obstruction.

The antipode does not change this conclusion. Multiplying the boundary line by the scalar sign ± 1 commutes with coordinate permutations, and it replaces the barycenter by $\pm\mathbf{1}_S/n$ without changing its denominator. Hence the obstruction is a finite equivariance/denominator obstruction, not a new $B\mathbb{Z}/2$ local-system class.

The checker `artifacts/reports/pass108-integral-equivariant-repair-section-check.json` verifies the invariant-lattice calculation, exact divisibility criterion, denominator obstruction, barycentric transition behavior under support inclusions, antipode independence, and overall PASS verdict.

The next task is to compute the rational transition $e_{S,T}s_{\mathrm{bar},S} - s_{\mathrm{bar},T}$ for support inclusions $S \subset T$, record its denominator and kernel class, and compare that bookkeeping with finite conductor and CRT denominator data.